A Comparative Study for Designing and Modeling Patch Antenna with Different Electromagnetic CAD Approaches — A Case Study

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Abstract— This paper presents a comparative study for designing and modeling patch antenna with different electromagnetic computer aided design (CAD) approaches as CST, HFSS, and FEKO. The patch antenna is an insert feed type with a copper patch and Roger RT/duroid substrate as a case study. The Wi-Fi operating frequency of 2.45 GHz is chosen for the resonator design. The effectiveness and limitations for each modeled approach for this case study are discussed in term of the antenna performance analysis.

1. INTRODUCTION

Recently, electromagnetic computer simulation modeling is considered one of the most efficient tools for designing and analyzing different microwave devices like antennas, filters, power dividers and so on. Within commercial software packages, researchers could study the behavior of their proposed design and they could optimize and modify it to meet their required before the fabrication of the final prototype model. Additionally, software tools provide many types of analysis such as 3-D analysis and visualization of current distribution which cannot be verified using experimental measurements [1, 2].

Designing and modeling of planner antennas such as microstrip patch antennas take the interest of many researchers in different fields of wireless communication systems because of its numerous advantages, including easy modeling, optimizing, and analyzing with different simulation tools, additionally they have good physical characteristics as low profile, light weight, simple structure, and can be easily intergraded with printed circuit board (PCB) microwave circuits [3–5].

All commercial simulation tools and approaches are based on solving Maxwell equation by different techniques In case of High Frequency Structural Simulator (HFSS), the finite element methods (FEM) is the main solver of Ansoft HFSS. FEKO is based on Integral Equations (IE) solved by Method of Moments (MoM). The Finite Difference Time Doman (FDTD) is used for Computer Simulation Technology (CST) in case of time domain transient solver [6]. Moreover there are various advantages of the above tools that they can be easily controlled by Math Works-MATLAB packages for optimization purposes [7].

In this case study, the designing and modeling of an insert-feed microstrip patch antenna will be presented. The proposed antenna is designed to resonant at frequency of 2.45 GHz. The antenna will be modeled and analyzed using different electromagnetic CAD software approaches as CST, HFSS, and FEKO. The performance of the antenna with each approach in terms of S-parameters, radiation patterns, voltage standing wave ratio (VSWR), impedance bandwidth and antenna gain will be discussed and compared.

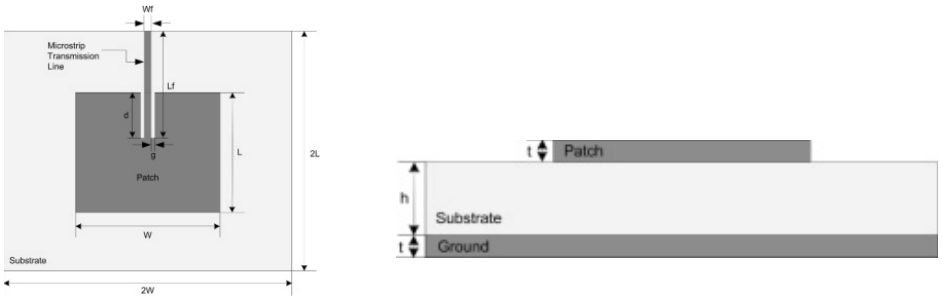
2. ANTENNA DESIGN ASPECTS

The proposed rectangular microstrip antenna configuration consist of copper patch on the Roger RT/duroid substrate with the finite copper ground sheet as shown in Figure 1. The design procedure conventionally explained in literatures [8, 9].

The specifications of the proposed antenna parameters are presented in Table 1 which will be used in the simulation models, CST, HFSS, and FEKO.

3. SIMULATION RESULTS

The performance of different electromagnetic simulation approaches for the proposed antenna measured in terms of the reflection coefficient S_{11} is illustrated in Figure 2. From this figure, it is clear that the antenna has a single bandwidth with resonant frequency of 2.45 GHz. CST approach



(a) (b)

Figure 1. Geometry of the insect-feed microstrip patch antenna. (a) Top view, (b) Front view.

Table 1. Specifications of the proposed antenna.

Parameters	Values
Operating frequency (f_r)	2.45 GHz
Dielectric constant (ϵ_r)	2.33
Dielectric loss tangent ($\tan\delta$)	0.0012
Substrate thickness (h)	0.787 mm
Patch thickness (t)	0.07 mm
Length of the patch (L)	39 mm
Width of the patch (W)	47 mm
Position of insert feed point (d)	14.8 mm
Gap between patch and feed line (g)	1 mm
Width of the microstrip feed line (W_f)	2.3 mm
Length of microstrip feed line (L_f)	30 mm

provides estimated resonant frequency sharply at 2.456 GHz correspond to −15.5 dB which differs from the design value by small error of 0.245% and the antenna bandwidth of 0.8% measured at −10 dB.

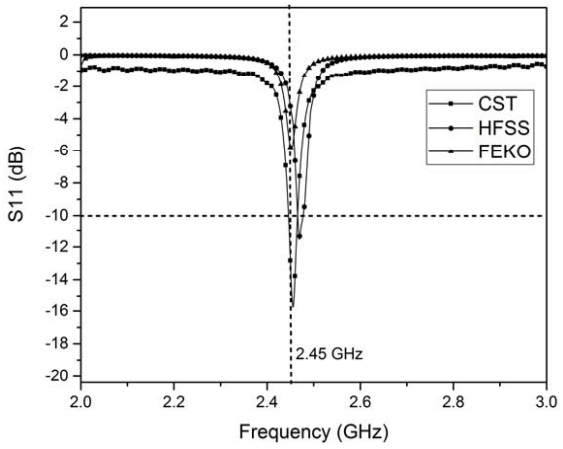


Figure 2. Reflection coefficient S_{11} for the proposed antenna with different simulation approaches.

In this approach, the simulated results of VSWR at the corresponding resonant frequency is 1.3 as shown in Table 2. Also, HFSS approach estimates the resonant frequency as 2.468 GHz at −13.45 dB with an error of 0.735% and the related bandwidth is 0.49% at −10 dB which is narrower than that of CST approach. The simulated VSWR at the corresponding resonant frequency is equal to 1.71 as shown in Table 2 which still in the acceptable range from 1.0 to 2.0. In case of the antenna modeled using FEKO software package, it is clear that this approach falls to reach −10 dB. Although it estimates resonant frequency of 2.452 GHz at −5.85 dB which in good agreement with

the designed value with error of 0.081%, it shows VSWR of 3.07 as shown in Table 2 which is greater than the required maximum value of 2.0. The simulation of S_{11}, impedance bandwidth and VSWR results for the proposed antenna using all the above mentioned approaches are listed in Table 2.

Table 2. Performance analysis of CST, HFSS, and FEKO.

Parameters	**CST**	**HFSS**	**FEKO**
Estimated Resonant Frequency, GHz	2.456	2.468	2.452
Error in Estimated Resonant Frequency, %	0.245	0.735	0.081
Impedance Bandwidth, % (Reference to −10 dB)	0.8	0.49	$S_{11} > -10$ dB
VSWR	1.3	1.71	3.07

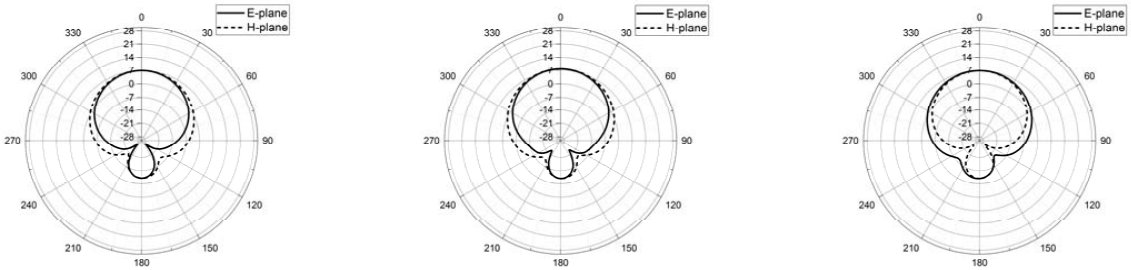


Figure 3. Radiation pattern for the proposed antenna with different simulation approaches at its estimated operating frequencies (a) CST, (b) HFSS, and (c) FEKO.

The simulated far-field radiation pattern in E-plane and H-plane for the proposed patch antenna at the resonant frequency of 2.45 GHz is depicted in Figure 3. For the three approaches, it is clear that the antennas' simulated radiation patterns are in the broad-side direction which is normal to the patch configuration. Furthermore, the simulated radiation performance has a slightly symmetric pattern in both E-plane and H-plane, which is a characteristic of patch antennas.

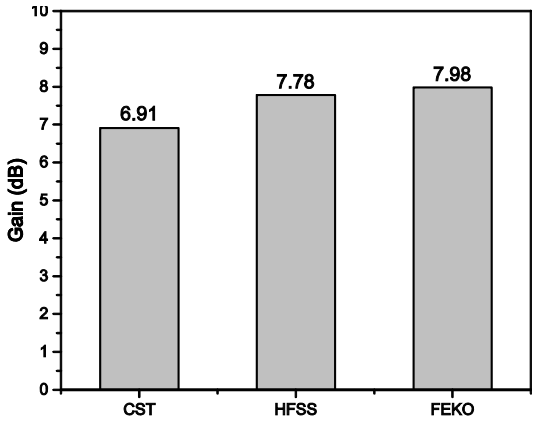


Figure 4. Maximum value of gains for different approaches at its estimated operating frequencies.

The variation of simulated gain for the proposed antenna accomplished by CST, HFSS, and FEKO approaches is illustrated in Figure 4. In this case study, Approaches for the proposed antenna models realize maximum gain of 6.91 dB, 7.78 dB, and 7.98 dB at the operating frequency for CST, HFSS, and FEKO respectively.

4. CONCLUSION

This comparative case study presents a designing and modeling of an insert-feed microstrip patch antenna. The proposed antenna is designed to resonant at frequency of 2.45 GHz. The antenna is

modeled and analyzed using different electromagnetic CAD software approaches as CST, HFSS, and FEKO. The effectiveness and limitations are discussed based on the performance of the proposed antenna models.

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